Title

Cross-Tissue Co-expression Analysis of PROX1 Reveals Conserved Association with Cell Identity Programs in Human Retina and Pancreas

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Abstract

The transcription factor PROX1 plays a critical role in cellular differentiation and tissue development and has been experimentally implicated as a regulator of regenerative capacity in retinal tissue. However, whether PROX1 exhibits consistent molecular associations across different human tissues remains unclear.

In this study, we perform a comparative single-cell RNA sequencing (scRNA-seq) analysis of human retina and pancreas datasets to characterize gene expression patterns associated with PROX1. Using correlation-based co-expression analysis, we identify gene programs linked to PROX1 activity in each tissue and assess overlap between them.

We find that PROX1 is associated with endothelial identity-related gene expression in the retina and with progenitor and ductal cell regulatory programs in the pancreas. A subset of shared co-expressed genes suggests potential conservation of underlying regulatory pathways related to cellular identity.

These findings indicate that PROX1 is consistently associated with gene networks involved in maintaining cell identity across tissues. While this study does not establish a causal role in regeneration, it provides computational evidence supporting further investigation into PROX1 as a candidate regulator of cellular plasticity.

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1. Introduction

Understanding the molecular mechanisms that regulate cellular identity and plasticity is central to advancing regenerative biology. The transcription factor PROX1 has been widely studied for its role in development, including lymphatic specification and organogenesis. Recent experimental studies have demonstrated that PROX1 can function as a barrier to regeneration in retinal tissue, where its inhibition enables reprogramming and neuronal regeneration.

Despite these findings, the extent to which PROX1 exhibits conserved molecular associations across tissues remains unclear. In particular, the pancreas represents a biologically and clinically relevant system where regeneration is limited and poorly understood.

This study aims to investigate whether PROX1 is associated with similar gene expression programs across retina and pancreas using single-cell transcriptomic data. Rather than testing regeneration directly, we focus on identifying patterns of co-expression that may reflect shared regulatory roles.

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2. Methods

2.1 Data Sources

- Retina: GSE135922 (human RPE/choroid scRNA-seq)
- Pancreas: Tabula Sapiens dataset (CELLxGENE Census)

2.2 Preprocessing

Data were processed using Scanpy:

- Quality control filtering
- Normalization and log transformation
- Dimensionality reduction (PCA, UMAP)
- Clustering (Leiden algorithm)

2.3 PROX1 Expression Analysis

- PROX1 expression was visualized across cells using UMAP and violin plots
- Expression patterns were examined across annotated cell populations

2.4 Co-expression Analysis

- Spearman correlation coefficients were computed between PROX1 and all other genes
- Genes were ranked by correlation strength within each tissue

2.5 Cross-Tissue Comparison

- Top-ranked co-expressed genes (top 50) were compared between tissues
- Shared and tissue-specific gene sets were identified

2.6 Pathway Enrichment

Gene sets were analyzed using gProfiler to identify associated biological processes.

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3. Results

3.1 PROX1 Expression Across Tissues

PROX1 expression was detected in subsets of cells in both datasets:

- Retina: ~16% of cells
- Pancreas: ~15% of cells

Expression patterns suggest tissue-specific localization within defined cellular populations.

3.2 Retina Co-expression Patterns

Top genes positively correlated with PROX1 include:

- PECAM1, LYVE1, FLT4, VEGFC, FOXC2

These genes are associated with:

endothelial and lymphatic identity-related processes

3.3 Pancreas Co-expression Patterns

Top genes positively correlated with PROX1 include:

- SOX9, KRT19, HNF1B, PDX1, NKX6-1, PTF1A

These genes are associated with:

progenitor, ductal, and developmental regulatory programs

3.4 Shared Gene Associations

Overlap analysis identified shared genes, including:

- HNF1B
- NRP2

These genes have known roles in:

developmental regulation and cellular identity across multiple tissues

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3.5 Pathway Enrichment

Enrichment analysis revealed:

- Retina: vascular and endothelial-related pathways
- Pancreas: developmental and transcriptional regulation pathways
- Shared: processes related to differentiation and cellular organization

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4. Discussion

This study demonstrates that PROX1 is associated with gene expression programs linked to cellular identity across both retina and pancreas. While the specific gene networks differ by tissue, the underlying pattern suggests a consistent association with identity-stabilizing processes.

These findings are consistent with prior experimental evidence indicating that PROX1 contributes to maintaining differentiated cellular states. In retinal systems, such stabilization has been shown to limit regenerative capacity. The observation that PROX1 is similarly associated with identity-related programs in the pancreas raises the possibility that it may play a comparable regulatory role in other tissues.

However, it is important to emphasize that this study is based on correlation analysis and does not establish causality. The relationship between PROX1 and regenerative potential in the pancreas remains to be experimentally tested.

5. Limitations

- Correlation does not imply causation
- Regenerative outcomes were not directly measured
- Dataset differences may introduce variability
- Cell-type annotations may affect interpretation

6. Future Directions

Future work should:

- Experimentally evaluate PROX1 modulation in pancreatic models
- Compare PROX1-high vs PROX1-low cell populations
- Extend analysis to additional tissues
- Integrate perturbation-based datasets to assess causal relationships

7. Conclusion

PROX1 is consistently associated with gene expression programs related to cellular identity across retina and pancreas. While these findings do not demonstrate a direct role in regeneration, they suggest that PROX1 may be a relevant candidate for further investigation in the context of cellular plasticity and regenerative biology.